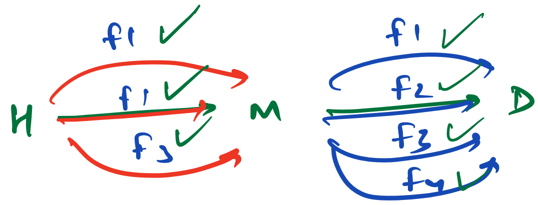
"Hello Everyone!"

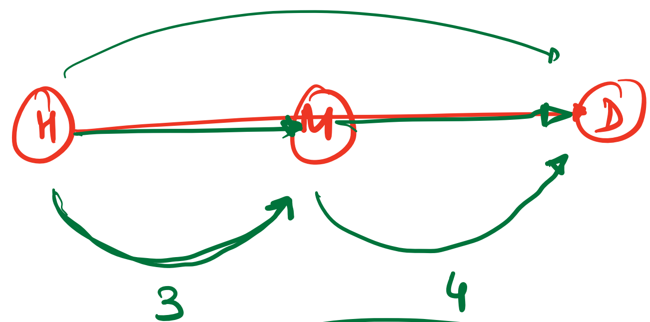
Maths 4: Combinatorics

(Q.1)



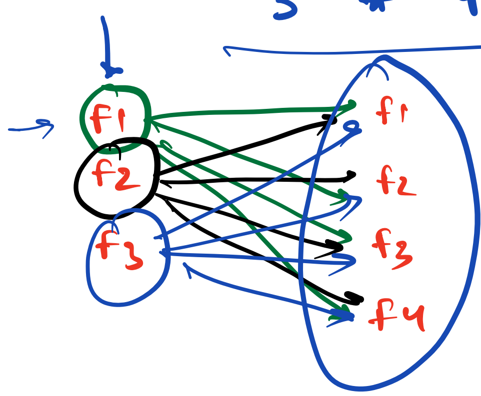
H → M

M → D



H → M **AND** M → D

$3 * 4 = \boxed{12 \text{ w}}\text{ays}$



12

(Q.1) # ways to do task A → n

ways to " B → m

① A and B both → n * m

② A or B → n + m

(Q.3) # ways $\underline{H \text{ to } M}$ $\underline{\underline{\text{or}}}$ $\underline{M \text{ to } D}$

$$\underbrace{3 + 4}$$

$\underline{(Q.2)}$

$$H \xrightarrow[2]{} N \xrightarrow[3]{} D$$

$\boxed{2 \times 3}$ ways

$H \to N$: f_1, f_2

$H \to D$: f_3, f_4, f_5

$\to$ $\underline{6 \text{ ways}}$

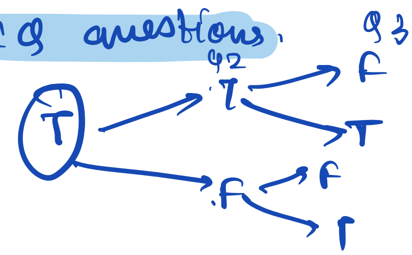
(Q.4)

In exam → 3 T/F Questions
→ 2 MCQ → 1 out of 4 options is correct.

(Q) → # ways to answer all the T/F question

OR

to answer all the MCQ questions.



All T/F Qustns

Q1 → T/F → 2
Q2 → T/F → 2
Q3 → T/F → 2

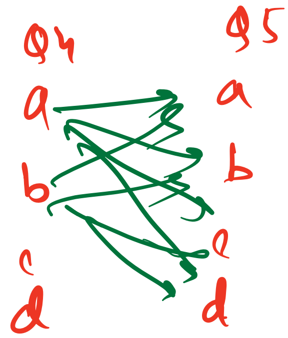
→ 2 * 2 * 2 = 8

All MCQ

Q4 → 4
*
Q5 → 4

OR
AND

4 * 4 → 16



$$\underline{8} + \underline{16} = 24$$

Q1 T/F   Q2 T/F $\longrightarrow$ 4

Q1   Q2

T   T

F   F

TT $\longrightarrow$

TF $\longrightarrow$

FT $\longrightarrow$

FF $\longrightarrow$

$\left.\right\}$ 4 ways

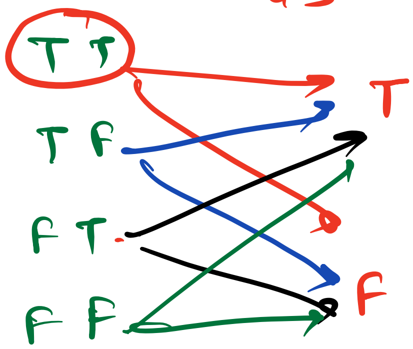
Q1   Q2   Q3

T/F   T/F   T/F

T   T   T

F   F   F

Q3



4×2

$$\underline{2 \times 2 \times 2} \longrightarrow \underline{\underline{8}}$$

Permutations → Ordered Arrangements

(Q) no. of permutations / no. of ways to arrange n distinct characters.

eg n = 3

a b c

"abc",
"acb",
"bac",
"bca",
"cab",
"cba".

= 6 ways

3

$3 * 2 * 1 = 6$

n = 4

a
b
c
d

$4 * 3 * 2 * 1 = 24$

$$\boxed{\# \text{ arrangements} \longrightarrow n!}$$

(Q.5) Find the # ways to arrange r out of n distinct characters.

eg:

$n = 3$

$r = 2$

$3 * 2 \longrightarrow \boxed{6}$

* **generalize**

arrange r out of n

r^{th} place

$$n * (n-1) * (n-2) * \cdots * (n-r+1)$$

$1 \longrightarrow n$

$2 \longrightarrow n-1$

$3 \longrightarrow n-2$

$r \longrightarrow n-(r-1) = \underline{n-r+1}$

$$\left(\boxed{n * (n-1) * (n-2) \cdots (n-r+1)} * \underbrace{(n-r)}_{(n-r)} * \underbrace{(n-r-1)}_{(n-r-1)} \cdots \underbrace{1}_{1}\right)$$

$$= \frac{n!}{(n-r) * (n-r-1) * (n-r-2) \cdots 1}$$

$$\boxed{{}^{n}P_{r} = \frac{n!}{(n-r)!}}$$

${}^{n}P_{r}$

arrange r out of n

$$5 \times \frac{10}{10} \times \frac{20}{20} \times \frac{39}{39}$$

$n \longrightarrow$

* Combinations → selection

Q) Find the no. of ways to select r out of n distinct characters.

eg :- $n = 3$, $r = 2$

a , b , c → {a, b}

{b, c}

{c, a}

→ 3

Cricket

3 players

selecting a team

MSD VK HP

1. {MSD, VK}
2. {HP, VK}
3. {MSD, HP}

→ Combinations

Batting order of 2 players

MSD, VK → diff from → VK, MSD

1. VK
2. MSD

1. MSD
2. VK

1. Selecting a Team → Combination
2. Batting order → $\boxed{\text{Permutation}}$

arrange

$n \longrightarrow n!$

n items

arrange any r items

① select any r items from n items

② arrange all the r items

AND

eg:- $n=4$, $r=2$

$\{a, b, c, d\}$

$\{a, b\}$ →	ab , ba
$\{a, c\}$ →	ac , ca
$\{a, d\}$ →	ad , da
$\{b, c\}$ →	bc , cb
$\{b, d\}$ →	bd , db
$\{c, d\}$ →	cd , dc

r →

$${}^{n}P_{r} = {}^{n}C_{r} * r!$$

$${}^{n}C_{r} = \frac{{}^{n}P_{r}}{r!}$$

$$\boxed{{}^{n}C_{r} = \frac{{}^{n}P_{r}}{r!}}$$

selecting r objects among n objects

$$^nP_r = {}^nC_r * r!$$

$$\boxed{^nC_r = \frac{^nP_r}{r!}}$$

$$= \frac{n!}{(n-r)!} * \frac{1}{r!}$$

$$\boxed{^nC_r = \frac{n!}{(n-r)! * (r)!}}$$

Properties :-

① # ways to select nothing from n distinct items → 1

→ $${}^{n}C_0 = \frac{\cancel{n!}}{\cancel{(n-0)!} * 0!} = \frac{1}{0!} = 1$$

$0! = 1$

DM AND KK AND Mukesh

$1 * 1 * 1 = 1$

(2) # ways to select all the n distinct items. → 1

$$ {}^{n}C_{n} = \frac{\cancel{n!}}{(n-n)!\,\cancel{n!}} = 1 $$

(3) [# ways to select $(n-r)$ items from n distinct items.]

$$ {}^{n}C_{r} = \frac{n!}{(n-r)!\,r!} $$

select r

$$ {}^{n}C_{(n-r)} = \frac{n!}{(n-(n-r))! * (n-r)!} $$

$r \leq n$

$$ = \frac{n!}{r! * (n-r)!} = \boxed{{}^{n}C_{r}} $$

ways to select any no. of the items out of n distinct items.

0, 1, 2, 3, ... n

$$\boxed{{}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n}$$

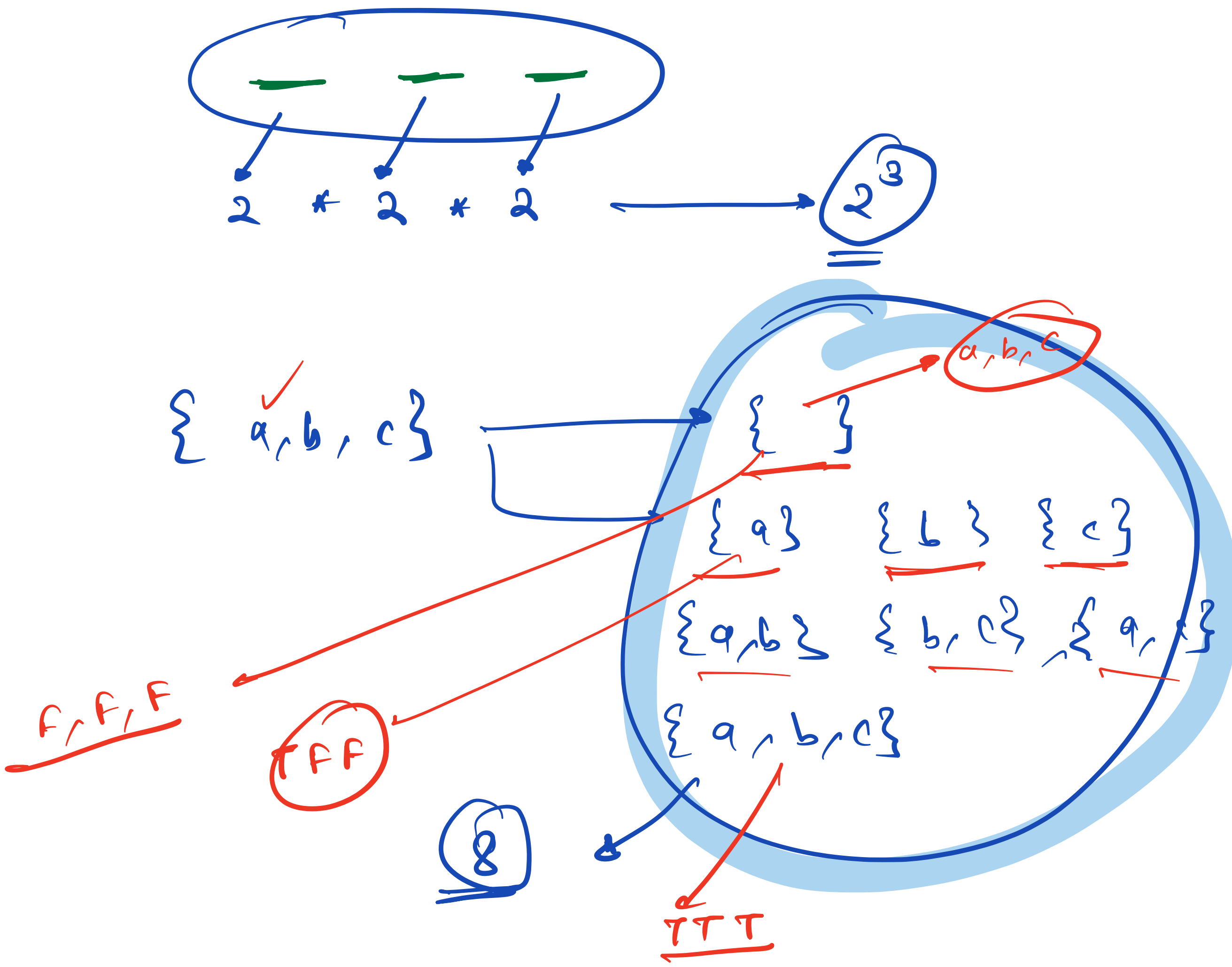
OR OR

$= 2^n$

Q1 Q2 Q3

T/F T/F T/F

2 2 2


2 * 2 * 2
2^3
{ a, b, c }
{ }
a, b, c
{ a } { b } { c }
{ a, b } { b, c } { a, c }
{ a, b, c }
F, F, F
TFF
8
TTT

P-5

$$^{n}C_{r} + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}$$

How to validate and tell in next lec!

10:32 → 10:40 → 10:42

(Q) ① Pascal Triangle for given N.

Part ② (with % M for all values)

$^0C_0$

$^1C_0$ $\quad$ $^1C_1$

$^2C_0$ $\quad$ $^2C_1$ $\quad$ $^2C_2$

$^3C_0$ $\quad$ $^3C_1$ $\quad$ $^3C_2$ $\quad$ $^3C_3$

$^4C_0$ $\quad$ $^4C_1$ $\quad$ $^4C_2$ $\quad$ $^4C_3$ $\quad$ $^4C_4$

(Q1) give first n rows of Pascal T (above)

(Q2) given nth row of PT.

---

till n^{th} row $\quad$ $\boxed{N=4}$

0	${}^0C_0$				
1	${}^1C_0$	${}^1C_1$			
2	${}^2C_0$	${}^2C_1$	${}^2C_2$		
3	${}^3C_0$	${}^3C_1$	${}^3C_2$	${}^3C_3$	
	${}^4C_0$	${}^4C_1$	${}^4C_2$	${}^4C_3$	${}^4C_4$

$\Rightarrow$

0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
4	1	4	6	4	1

$${}^2C_1 = \frac{2!}{(2-1)!\,1!} = \frac{2}{1\times 1} = 2$$

$${}^4C_2 = \frac{4!}{(4-2)!\times 2!} = \frac{4!}{2!\times 2!} = 6$$

$$\% M \longrightarrow [0 \qquad M-1]$$

Concept :-

$3! = 6$

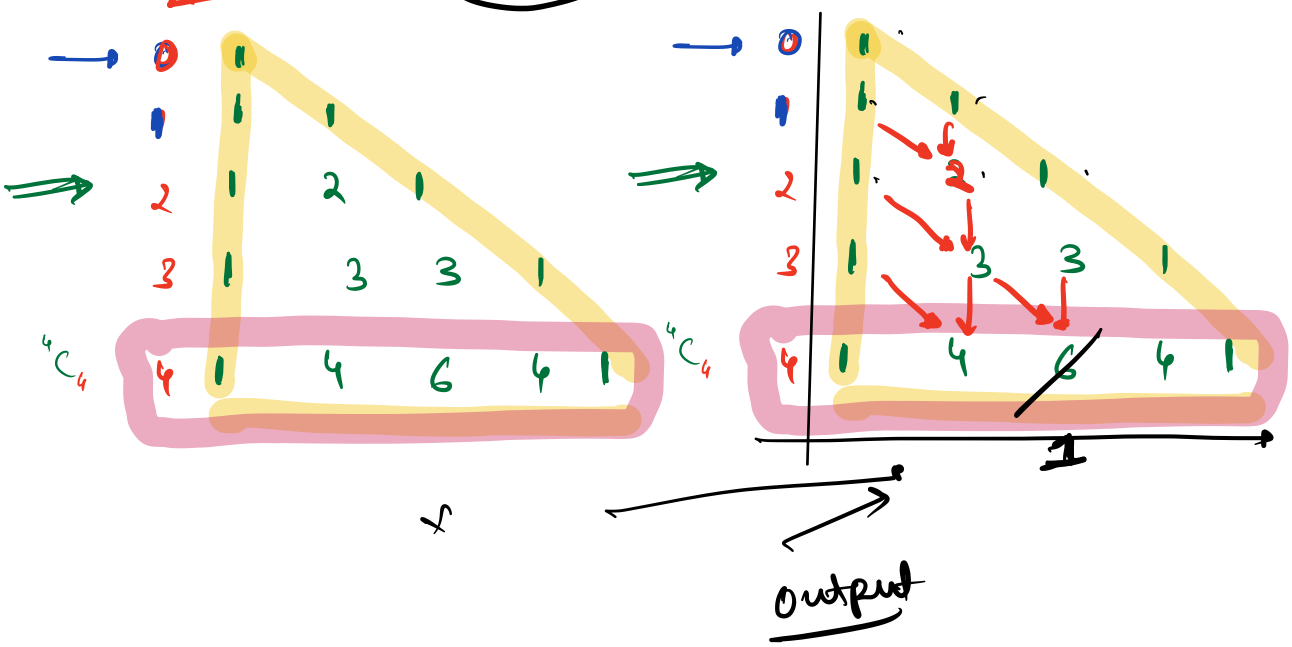
$5! = 120$

$10! = 3628800$

$20! = \ldots\ldots$ 19 digit number

Part 2

$\boxed{N=4}$ $M=5$ $\boxed{N=4}$



* Part-2 code

Obs :

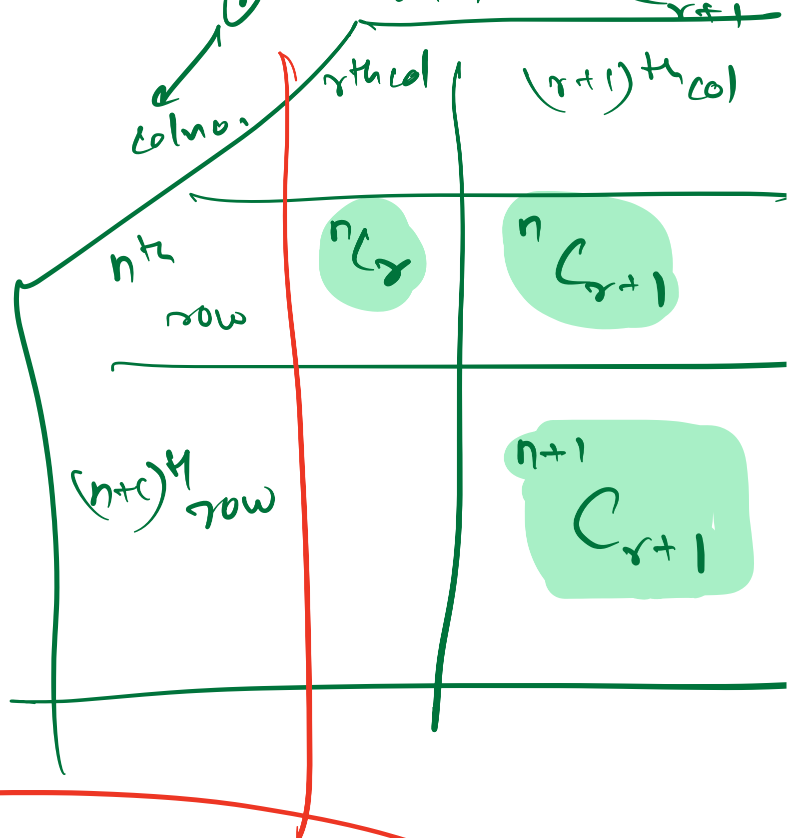
$^nC_0 = 1$

$^nC_n = 1$

row no

$$^{n}C_{r} + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}$$

col no.



i

row

^iC_j

col

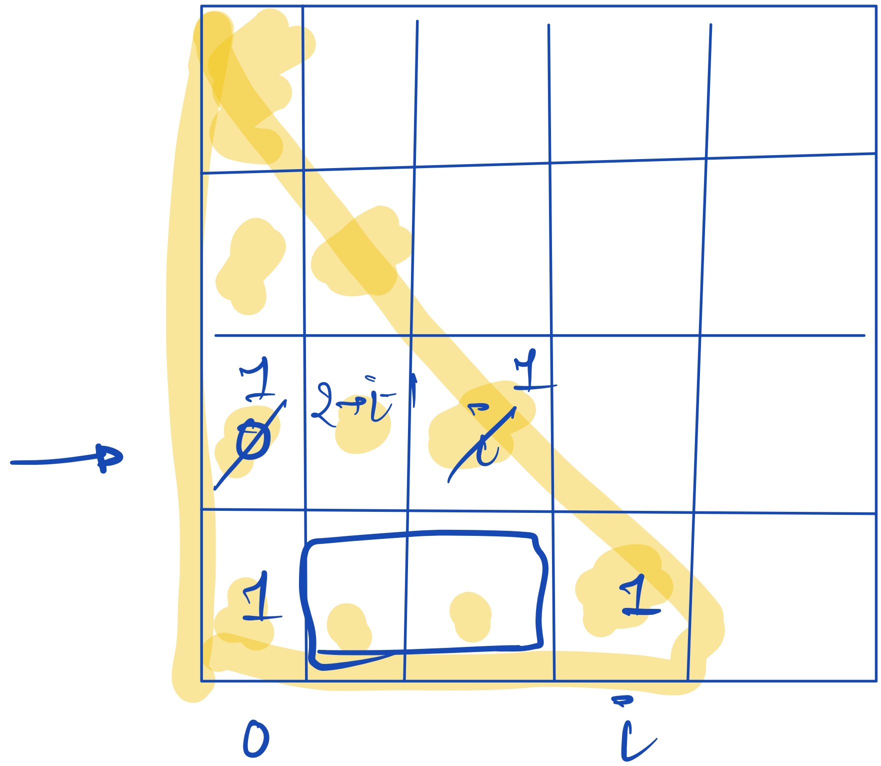
$$^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$$

$n \to i-1$

$r \to j-1$

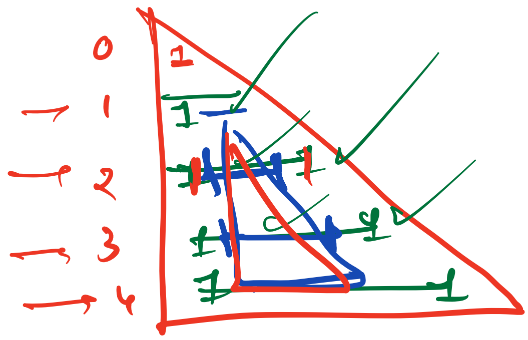
$$^{i-1}C_{j-1} + {}^{i-1}C_j = {}^{i}C_j$$

$$\boxed{c[i][j]\,\%\,m = (c[i-1][j-1] + c[i-1][j])\,\%\,m}$$

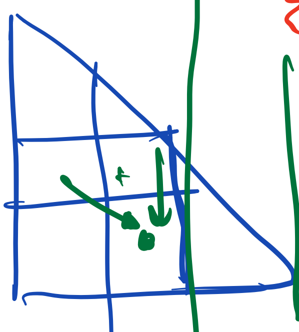


$$(a+b)\,\%\,m = (a\,\%\,m + b\,\%\,m)\,\%\,m$$

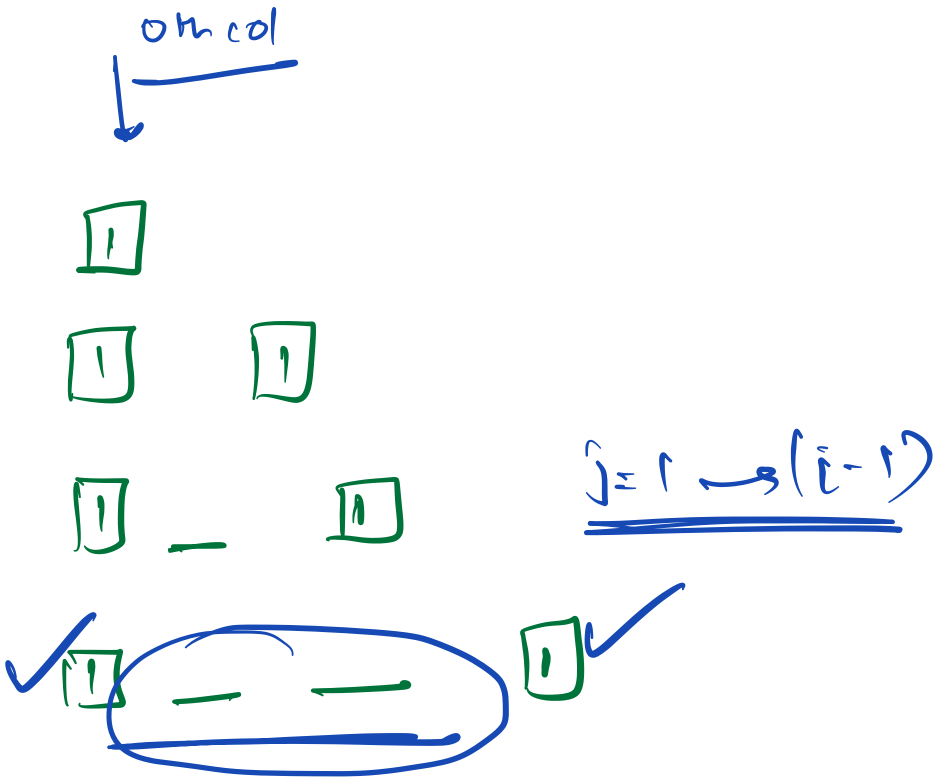
Pseudo Code :

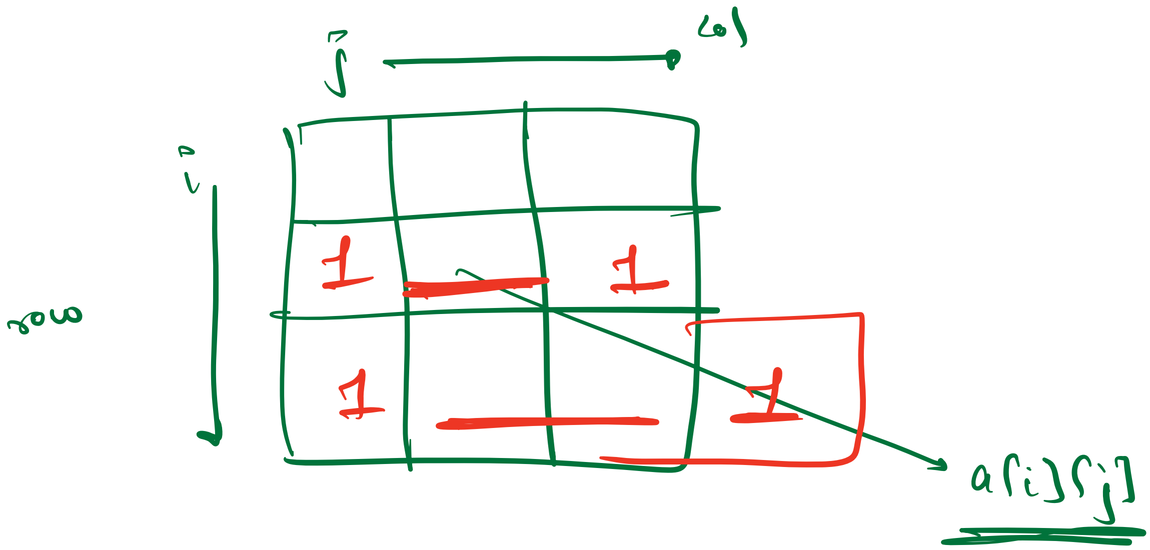


```
print(1)
print("\n"
for i : 1 → N        // rows
{  c[i][0] = 1
   c[i][i] = 1
   print(c[i][0])
   // fill the remaining values for every row
   for j : 1 → (i-1)   → cols
   {
      c[i][j] = (c[i-1][j-1] + c[i-1][j]) % m
      print( c[i][j])
   }
   print (c[i][i])
   print ("\n")
}
```



0 based indexing


on col
j=1 → (i-1)


j
col
i
row
1
1
1
1
a[i][j]

(Q) find ${}^{n}C_r \% p$, P → prime number

eg:-

$n = 5$
$r = 2$
$P = 7$

$${}^{5}C_2 = 10$$

$${}^{5}C_2 \% 7 = 10 \% 7 = 3$$

Ans

$${}^{n}C_r = \left(\frac{n!}{(n-r)! * r!} \% P \right)$$

${}^{n}C_r \longrightarrow$ nf $\longrightarrow n!$

nrf $\longrightarrow (n-r)!$

rf $\longrightarrow r!$

$$(x^{-1} \% P)$$

$$n! \% P * (n-r)^{-1} \% P$$

$$\left(\frac{n!}{(n-r)!\,r!}\right) \% P = \left((n! \% P) * \underline{((n-r)\% P)^{-1}} * \underline{(r! \% P)^{-1}} \right) \% P$$

$$\frac{1}{a} \rightarrow a^{-1}$$

inverse modulo of $(n-r)$ with p.

$$\boxed{x^{-1} \pmod{p} = x^{(p-2)} \% P}$$

prime no.

H.W : Find Time Complexity to find $^{n}C_{r} \% P$

prime

TC

$n! \longrightarrow O(n)$

$(n-r)! \longrightarrow O(n)$

$r! \longrightarrow O(n) \longrightarrow$ worst case

$r <= n$

5!

for i : 1 ⟶ n

fact = fact * i

$n! \longrightarrow$

fact = 1

for i : 1 ⟶ n

fact = fact * i

$\longrightarrow O(n)$

Doubts